26 October 2026

Editors
Journal of Physics A: Mathematical and Theoretical

Dear Editors,

Please consider “The Cost of Reproducing Learning Trajectories: Sharp Resource Bounds for Adaptive Resistor Networks” as a Paper.

An instantaneous electrical equivalent need not preserve a network's response to further training. This article determines how many trainable conductances are necessary and sufficient to reproduce the complete family of learning trajectories of N parallel three-ray stars. Source and replacement obey the same conductance-weighted gradient law, target class, physical clock and initial conductance budget, and retain no auxiliary memory. On a fixed finite horizon, for sufficiently small uniform tolerance delta, the worst-case minimum edge count has order min{N, [1 + log(1/delta)]^2}. Exact reproduction requires 3N edges in the worst case, also below a positive N-dependent tolerance.

An error estimate for a chosen reduction scheme leaves open whether another physical topology could reproduce the trajectories with fewer elements. Our lower bound addresses that question across all admissible passive replacement graphs and arbitrary, possibly discontinuous preparation. A fixed star ensemble with one smooth, strictly positive initializer attains the matching upper order. The physical constraints enter both bounds: conductances must remain positive, and the replacement evolves under the same learning law and physical clock, driven by its own output residual. The proof combines topology-uniform sensitivity, distinguishability by finite training programs, and a constructive realization of analytic response features in trainable conductances.

The work is intended for readers in mathematical physics, nonlinear dynamics and networks with physical learning. It preserves complete proofs in five appendices. Its bounds are asymptotic: it does not claim practical hardware savings, an experimental implementation, or that a device solving the same final tasks must reproduce the same trajectories. Earlier related work on normalization and learning-history memory addresses different preservation problems, as explained in the Discussion.

A preprint and its document sources are available at https://doi.org/10.5281/zenodo.23655709. The requested route is subscription-access publication; Gold Open Access and paid services are not requested.

Thank you for considering the manuscript.

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